

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284059

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

LU; Yu et al.

SYSTEM FOR PROTECTING TELECOMMUNICATIONS CABLE SPLICE ARRANGEMENTS

Abstract

A splice arrangement including a splice protection housing that defines a passage that extends lengthwise through the splice protection housing from a first end of the splice protection housing to an opposite second end of the splice protection housing. The splice arrangement may also include a first fiber optic cable and a second fiber optic cable. First optical fibers of the first fiber optic cable and second optical fibers of the second fiber optic cable may be coupled together at a splice location positioned within the passage of the splice protection housing.

Inventors:	LU; Yu (Eden Prairie, MN), NHEP; Ponharith (Savage, MN), CARLSON; Scott L. (Bloomington, MN), BENDALE; Chinmay M. (Eden Prairie, MN), GONZALEZ BATISTA; Jaime (Prior Lake, MN), CANZONIERI; Paige E. (Brooklyn Park, MN)
Applicant:	CommScope Technologies LLC (Hickory, NC)
Family ID:	86059587
Assignee:	CommScope Technologies LLC (Hickory, NC)
Appl. No.:	18/702581
Filed (or PCT Filed):	October 18, 2022
PCT No.:	PCT/US2022/047040

Related U.S. Application Data

us-provisional-application US 63262648 20211018

Publication Classification

Int. Cl.: G02B6/255 (20060101)

U.S. Cl.:

CPC G02B6/2558 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is being filed on Oct. 18, 2022 as a PCT International Patent Application and claims the benefit of U.S. Patent Application Ser. No. 63/262,648, filed on Oct. 18, 2021, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Optical fiber communications systems are extensively used in the telecommunications industry due to their large information carrying capacity, their virtually noise-free performance and the long span distances achievable before regeneration and amplification of the signal is required. In many communication systems employing optical fibers, splice locations are provided where optical fibers are coupled together. Often these splice locations are located remote from central offices, for example, at customer premises or areas serviced by local area networks.

[0003] At each splice location, optical fibers must be separated from outer protective cable components for splicing and termination. However, removal of optical fibers from their outer protective cabling components makes the optical fibers vulnerable to damage. Protection of splice locations after splicing is an important consideration. One example of a telecommunication splice arrangement is shown in U.S. Patent Publication No. 2020/0348469, the disclosure of which is incorporated herein by reference.

SUMMARY

[0004] One aspect of the disclosure relates to a system for protecting a splice of telecommunications cable. The system can include a splice protection housing having a length that extends between a first end and a second end. The splice protection housing defines a fiber routing and splice containment passage the extends between a first end to an opposite, second end of the splice protection housing. The splice protection housing further includes first and second anchoring pockets at the first and second ends, respectively. An adhesive injection port is defined by the splice protection housing and in communication with the splice containment passage is positioned between adhesive vent ports positioned toward the first and second ends that are also defined by the splice protection housing.

[0005] The system may also include a first fiber optic cable. The first fiber optic cable may include a first outer jacket that contains first optical fibers and a first buffer tube and first strength members. The first fiber optic cable may be anchored to the first anchoring pocket.

[0006] The system may further include a second fiber optic cable. The second fiber optic cable may include a second outer jacket that contains second optical fibers and a second buffer tube and second strength members. The second fiber optic cable may be anchored to the second end anchoring pocket.

[0007] The first optical fibers and the second optical fibers may be coupled together at a splice location positioned within the splice containment passage. The splice protection housing may surround and support the splice location to provide structural reinforcement thereto.

[0008] The first and second anchoring pockets may include major and minor dimensions that are transposed 90 degrees from major and minor dimensions of the splice containment passage.

[0009] A variety of additional aspects will be set forth in the description that follows. The aspects

relate to individual features and to combinations of features. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the embodiments disclosed herein are based.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings, which are incorporated in and constitute a part of the description, illustrate several aspects of the present disclosure. A brief description of the drawings is as follows:

[0011] FIG. 1 is a perspective view of one embodiment of a system for protecting a splice of telecommunication cables including a splice protection housing in accordance with principles of the present disclosure;

[0012] FIG. 1A is a section view of the system for protecting a splice of telecommunication cables of FIG. 1, showing a splice containment passage;

[0013] FIG. 2 is a top view of the system for protecting a splice of telecommunication cables of FIG. 1;

[0014] FIG. 3 is a side view of the system for protecting a splice of telecommunication cables of FIG. 1;

[0015] FIG. 4 is a bottom view of the system for protecting a splice of telecommunication cables of FIG. 1;

[0016] FIG. 5 is a front view of the system for protecting a splice of telecommunication cables of FIG. 1;

[0017] FIG. 6 is an exploded perspective view of the system for protecting a splice of telecommunication cables of FIG. 1;

[0018] FIG. 7 is an exploded side view of the system for protecting a splice of telecommunication cables of FIG. 1;

[0019] FIG. 8 is a perspective view of the main body of the splice protection housing shown in FIG. 1;

[0020] FIG. 9 is a top view of the main body of the splice protection housing shown in FIG. 1;

[0021] FIG. 10 is a side view of the main body of the splice protection housing shown in FIG. 1;

[0022] FIG. 11 is a front view of the main body of the splice protection housing shown in FIG. 1;

[0023] FIG. 12 is a perspective view of the cover of the splice protection housing shown in FIG. 1;

[0024] FIG. 13 is a bottom view of the cover of the splice protection housing shown in FIG. 1;

[0025] FIG. 14 is a top view of the cover of the splice protection housing shown in FIG. 1;

[0026] FIG. 15 is a front view of the cover of the splice protection housing shown in FIG. 1; and

[0027] FIG. 16 is a perspective view of one embodiment of a splice protection housing in accordance with the principles of the present disclosure.

[0028] FIG. 17 is a perspective view of an embodiment of a splice protection housing with a fiber optic cable pre-terminated with a connection in accordance with the principles of the present disclosure.

[0029] FIG. 18 is a perspective view of another embodiment of a splice protection housing in accordance with the principles of the present disclosure.

DETAILED DESCRIPTION

[0030] Aspects of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings, in which illustrative embodiments are shown. This disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be

thorough and complete, and will fully convey the scope of the invention to those skilled in the art. [0031] Splice protections are often used in the field to join two lengths of fiber optic cables together. For example, a splice protection may be used to splice a first cable to a second cable to provide an extended cable length or to provide a cable repair. For the present application, the splice protection is used to optically connect a fiber optic connector (e.g., a multi-fiber optical connector such as a hardened multi-fiber optical connector (HMFOC) to a longer cable). In certain examples, the fiber optic connector is pre-terminated at the end of a relatively short length of fiber-optic cable to form a connectorized tether. The non-connectorized end of the connectorized tether is spliced to a longer length of cable at a joint to provide a connector as cable assembly such as a connector eyes to drop cable assembly.

[0032] Referring to FIGS. **1-17**, one example of a splice for telecommunication cable **100** according to the disclosure is shown. The splice **100** may include a splice protection housing **102**, a first fiber optic cable **104** (e.g., a single fiber or multi-fiber cable), and a second fiber optic cable **106** (e.g., a single fiber or multi-fiber cable). The first and second fiber optic cables **104**, **106** may each have on the order of 12 to 48 fibers; however, alternative implementations may include fewer or more fibers. In certain examples, and as shown in FIG. **17**, first fiber optic cable **104** is pre-terminated with a connector **107**.

[0033] In certain embodiments, first and second fiber optic cables **104**, **106** do not have round cross sections. Instead, each has an elongate cross-sectional profile having a major axis A and a minor axis B that are perpendicular relative to one another. Each cable **104**, **106** includes an outer jacket **108**, **110**, which defines the outer cross-sectional profile, a buffer tube **112**, **114** within the jacket which contains the optical fibers **116**, **118**, and two strength members **120**, **122** aligned along the major axis on opposite sides of the buffer tube. In one example, the strength members each has a construction including a glass fiber reinforced polymer (e.g., an epoxy reinforced by glass rovings).

[0034] In the example shown, each of the cables **104**, **106** includes a plurality of optical fibers **116**, **118**. The optical fibers **116**, **118** of each of the cables **104**, **106** are spliced together at a splice location **150** within the splice protection housing. In a preferred example, the optical fibers **116**, **118** at the splice location **150** are arranged in a planar configuration in which the optical fibers are aligned along a plane. In certain examples, the splice location **150** can be protected by a film or other reinforcing structure bonded to the optical fibers **116**, **118** that extends across the splice location. A cross-sectional profile of the splice location **150** can have a major dimension and a minor dimension because the splice location can be arranged in a planar configuration. In certain examples, the optical fibers **116**, **118** of one or both of the cables **104**, **106** can be ribbonized (bonded together by a matrix material such as acrylate) adjacent the splice location **150** to assist in maintaining an order and positioning of the optical fibers prior to splicing.

[0035] Splice protection housing **102** has an elongate configuration and includes a length **130** that extends between opposite first **132** and second ends **134**. In certain embodiments, the splice protection housing **102** defines first and second cable anchoring pockets **136**, **138** respectively adjacent the first and second ends. The splice protection housing **102** also defines a fiber routing and splice containment passage **140** that extends through the length **130** of the splice protection housing between the first and second cable anchoring pockets **136**, **138**. The fiber routing and splice containment passage **140** is in communication with the first and second cable anchoring pockets **136**, **138**.

[0036] Splice protection housing **102** further includes a main body **160** and a cover **162** that enclose an interior passage **164** when they are joined together. Fiber routing and splice containment passage **140** and first and second cable anchoring pockets **136**, **138** are defined by interior passage **164**. In one example cover **162** can be secured to main body **160** by a snap-fit connection. Cover **162** may include a plurality of securing tabs **166** positioned along the length of the cover that selectively engage securing channels **168** that extend along the length of main body **160**. In another example, cover **162** and main body **160** may be secured to each other by an arrangement of latches.

In one example, latches are provided at the first and second cable anchoring pockets **136**, **138** as well as at at least one intermediate location between the first and second cable anchoring pockets. [0037] Splice protection housing **102** may be formed from any suitable material with plastic such as but not limited to polypropylene being a desirable material. It will be appreciated that the splice protection housing **102** can be made from, plastics, ceramics, acrylics, rubber, combinations thereof, or any other solid material.

[0038] An adhesive injection port **142** is defined by the splice protection housing **102** in communication with the fiber routing and splice containment passage **140**. In one example, the adhesive injection port **142** is located at a midpoint between the first and second ends of the splice protection housing. First and second adhesive vent ports **144**, **146** are defined by the splice protection housing **102** respectively in communication with the first and second cable anchoring pockets **136**, **138**. The adhesive injection port **142** as well as the first and second vent ports **144**, **146** are all accessible at an exterior of the splice protection housing **102**.

[0039] By injecting an adhesive such as epoxy into the splice protection housing through the adhesive injection port, the fiber routing and splice containment passage **140** as well as the first and second cable anchoring pockets **136**, **138** can be filled with adhesive in one shot. It will be appreciated that the adhesive is preferably continuously injected until adhesive begins to bleed through both of the first and second vent ports **144**, **146**.

[0040] In one example, the cover **162** can define the adhesive injection port **142** as well as the first and second vent ports **144**, **146**. In another example, the main body **160** can define the adhesive injection port **142** as well as the first and second vent ports **144**, **146**. In yet another example, the adhesive injection port **142** and first and second vent ports **144**, **146** may be defined on a combination of the main body **160** and cover **162**. In some embodiments, multiple adhesive injection ports **142** can be included and positioned in any suitable location on the splice protection housing **102** without departing from the principles of this disclosure. In some examples, first and second vent ports **144**, **146** may not be present, and clearance between the housing and the cable provides venting.

[0041] As shown in FIG. 5, the first and second cable anchoring pockets **136**, **138** each have a cross-sectional profile including a major dimension A and a minor dimension B. Similarly, and as shown in FIG. 1A, the fiber routing and splice containment passage **140** has a cross-sectional profile including a major dimension C and a minor dimension D. The major dimension C and the minor dimension D of the cross-sectional profile of the fiber routing and splice containment passage **140** are transposed 90° relative to the major dimensions A and the minor dimensions B of the first and second cable anchoring pockets **136**, **138**. In some examples, the cross-sectional profile of the fiber routing and splice containment passage **140** has an area that is less than 25%, or less than 20%, less than 15% or less than 10% of a total cross-sectional profile of the splice protection housing **102**.

[0042] In one example, shown in detail in FIGS. 1A, 8, and 12, the fiber routing and splice containment passage **140** can be defined by inner walls **170**, **172** of the splice protection housing **102**. In one example, the inner walls **170**, **172** can be generally parallel to one another. The inner walls **170**, **172** are separated from exterior defining portions of the cable joint housing by first and second open regions **174**, **176** that extend along the length of the cable joint housing on opposite sides of the fiber routing and splice containment passage. The open regions can be subdivided into a plurality of open chambers **178** by reinforcing ribs **180** that extend between the inner walls **170**, **172** and the exterior defining portions of the cable joint housing. In certain examples, relatively small amounts of excess adhesive injected into the fiber routing and splice containment passage **140** may overflow into the first and second open regions **174**, **176**.

[0043] The first and second cable anchoring pockets **136**, **138** include features for assisting in effectively bonding the cables **104**, **106** within their respective pockets and for assisting in retaining the cover **162** on the main body **160** of the splice protection housing **102**. For example, the first and

second cable anchoring pockets **136**, **138** include adhesive retention ribs **124** between which the adhesive flows when the adhesive is injected into the cable joint housing. Once the adhesive cures, mechanical interference between the adhesive and the adhesive retention ribs **124** provides an interlock that prevents the adhesive from being axially withdrawn from the ends of the splice protection housing **102**. At the first and second cable anchoring pockets, the seam **126** between the cover **162** and the main body **160** of the cable joint housing is defined with a predefined gap **128** that allows the adhesive to flow into the gap to provide bonding at the seam between the cover and the main body of the splice protection housing **102**.

[0044] To install the splice protection housing **102** over the first and second cables **104**, **106**, the first and second cables are first prepared. For example, the cables **104**, **106** are so prepared by initially stripping off a section of the cable outer jacket **108**, **110** to expose the buffer tubes **112**, **114** and the strength members **120**, **122**. The buffer tubes **112**, **114** are trimmed to expose the optical fibers **116**, **118** of the cables **104**, **106**. Preferably, a sufficient length of optical fibers **116**, **118** is exposed to allow for performing a mass fusion splice between the optical fibers and to provide sufficient length of optical fiber through the splice protection housing **102**. The strength members **120**, **122** are also trimmed such that relatively short lengths of the strength members project beyond ends of the outer jackets **108**, **110** of the cables **104**, **106**. As previously described, once the cables **104**, **106** have been prepared and the optical fibers **116**, **118** have been spliced together, the optical fibers can be protected by a structure such as a laminate film or other structure.

[0045] Next, the splice location **150** of the optical fibers **116**, **118** is positioned within the fiber routing and splice containment passage **140** and the prepared ends of the fiber optic cables **104**, **106** are positioned within the respective cable anchoring pockets **136**, **138**. The splice location **150** is preferably positioned at a central location along the length of the main body **160** of splice protection housing **102**. The cover **162** is then mounted on the main body **160** of the splice protection housing **102** to enclose the fiber routing and splice containment passage **140** as well as to enclose the cable anchoring pockets **136**, **138**. An adhesive such as epoxy is then injected into the centrally located adhesive injection port **142**. Injection of the adhesive continues until the fiber routing and splice containment passage **140** and the cable anchoring pockets **136**, **138** are filled with adhesive. Adhesive flowing from both of the adhesive vent ports **144**, **146** provides an indication that the fiber routing and splice containment passage **140** as well as the cable anchoring pockets **136**, **138** are filled with adhesive. Once the adhesive has cured, the splice protection housing **102** installation processes is complete.

[0046] FIG. **16** shows an alternative embodiment of a splice protection housing **200** in accordance with the principles of this disclosure. Splice protection housing **200** is similar in many ways to the embodiment described above but rather than a two-piece construction, this embodiment is made of four parts. In the embodiment shown, splice protection housing **200** includes a main body **202**, a cover **204**, and two end caps **206**, **208**. In one example, cover **204** is secured to main body **202** by the two end caps **206**, **208** rather than by snap-fit as in the prior embodiment. Cover **204** includes end tabs **210** that selectively engage end caps **206**, **208** when splice protection housing **200** is assembled. In certain embodiments, cover **204** may be snap-fit onto main body **202**, with end caps **206**, **208** further securing the cover to the main body. When end caps **206**, **208** secure cover **204** to main body **202**, first and second cable anchoring pockets **236**, **238** are formed at either end of splice protection housing **202** by a combination of cable anchoring tabs **232**, **234** that are integrally formed in main body **202** and end caps **206**, **208**.

[0047] Splice protection housing **200** also includes a fiber routing and splice containment passage **240** that has a cross-sectional profile including a major dimension and a minor dimension. The major dimension and the minor dimension of the cross-sectional profile of the fiber routing and splice containment passage **240** are transposed 90° relative to a major dimension and minor dimension of the first and second cable anchoring pockets **236**, **238**.

[0048] In one example, splice protection housing **200** also includes an adhesive injection port **242**

and first and second adhesive vent ports **244, 246**. The adhesive injection port **242** is located at a midpoint between the first and second ends of the splice protection housing, is defined by the cover **204**, and is in communication with fiber routing and splice containment passage **240**. First and second adhesive vent ports **244, 246** are defined by the end caps **206, 208** respectively in communication with the first and second cable anchoring pockets **236, 238**. The adhesive injection port **242** as well as the first and second vent ports **244, 246** are all accessible at an exterior of the splice protection housing **200**.

[0049] FIG. **18** shows an alternative embodiment of a splice protection housing **300** in accordance with the principles of this disclosure. Splice protection housing **300** is similar in many ways to the embodiments described above but rather than filling the length of the housing with adhesive, adhesive is restricted to the ends **332, 334**.

[0050] In the embodiment shown, splice protection housing **300** includes a main body **302** and a cover **304**. In one example, cover **304** is secured to main body **302** by a snap-fit. When cover **304** is secured to main body **302**, first and second cable anchoring pockets **336, 338** are formed at either end of splice protection housing **302**. Splice protection housing **300** also includes a fiber routing and splice containment passage **340** that has a cross-sectional profile including a major dimension and a minor dimension. The major dimension and the minor dimension of the cross-sectional profile of the fiber routing and splice containment passage **340** are transposed 90° relative to a major dimension and minor dimension of the first and second cable anchoring pockets **336, 338**.

[0051] First and second adhesive injection ports **344, 346** are defined by the end **332, 334** of cover **304**, each respectively in communication with the first and second cable anchoring pockets **336, 338**. The first and second adhesive injection ports **344, 346** are each accessible at an exterior of the splice protection housing **300**. First and second adhesive injection sites **344, 346** are each provided with an associated first and second recessed groove **380, 382**, as well as being in communication with first and second cable anchoring pockets **336, 338**.

[0052] Adhesive injected into first and second adhesive injection ports **344, 346** is prevented from running into the central portion of splice containment passage **340** by barriers **384**. Barriers **384** permit fibers to be routed through but block adhesive from passing. In embodiments, barriers **384** are each formed with a body portion associated with main body **304** and a cover portion associated with cover **302**. When main body **304** and cover **302** are mated together, the body portion and cover portion of each barrier is likewise mated to form a complete barrier. Example embodiment **300** depicts two barriers between a central region of the splice containment **340** and each of first and second cable anchoring pockets **336, 338**, but other numbers of barriers are envisioned and supported. In embodiments, at least one barrier between a central region of splice containment passage **340** and each of first and second cable anchoring pockets **336, 338** may be desirable. Example embodiment **300** may be advantageous for bonding cable and short sections of fibers with adhesive while leaving the central splice section free of adhesive.

[0053] The various examples described above are provided by way of illustration only and should not be construed to limit the scope of the present disclosure. Those skilled in the art will readily recognize various modifications and changes that may be made without following the example examples and applications illustrated and described herein, and without departing from the true spirit and scope of the present disclosure.

Claims

1. A system for protecting a splice of telecommunications cable, the system comprising; a splice protection housing having a length that extends between a first end and a second end; the splice protection housing defining a fiber routing and splice containment passage extending between the first and second ends; first and second cable anchoring pockets respectively at the first and second ends; the splice protection housing defining a first adhesive vent port in communication with the

first cable anchoring pocket and a second adhesive vent port in communication with the second cable anchoring pocket; the splice protection housing defining an adhesive injection port positioned between the first and second adhesive vent ports and in communication with the fiber routing and splice containment passage; a first length of fiber optic cable including at least one first optical fiber; a second length of fiber optic cable including at least one second optical fiber; the at least one optical fiber spliced to the at least one second optical fiber at a splice location; the splice location disposed within the fiber routing and splice containment passage; the first length of optic cable secured within the first cable anchoring pocket; and the second length of optic cable secured within the second cable anchoring pocket.

2. The system of claim 1, wherein the first length of fiber optic cable is terminated by a connector.

3. The system of claim 1, wherein the adhesive injection port is centrally located between the first and second ends of the splice protection housing.

4. The system of claim 1, wherein the splice protection housing comprises a main body and a cover.

5. The system of claim 4, wherein the cover and main body are secured to each other by a snap-fit.

6. The system of claim 5, wherein the adhesive injection port and adhesive vent ports are defined by the cover.

7. The system of claim 1, wherein the first and second cable anchoring pockets have a first major dimension and a first minor dimension, the first major dimension greater than the first minor dimension.

8. The system of claim 7, wherein the fiber routing and splice containment passage has a second major dimension and a second minor dimension, the second major dimension greater than the second minor dimension.

9. The system of claim 8, wherein the first major and minor dimensions are transposed 90° relative to the second major and minor dimension.

10. The system of claim 1, wherein the fiber routing and splice containment passage is defined by two inner walls that extend between the cable anchoring pockets.

11. The system of claim 2, further comprising an adhesive material contained within the splice protection housing for encapsulating the first optical fibers and the second optical fibers.

12. The system of claim 1, wherein the splice protection housing is factory or field installable.

13. A splice protection housing comprising; a length that extends between a first end and a second end; first and second cable anchoring pockets respectively at the first and second ends; a fiber routing and splice containment passage extending between the first and second cable anchoring pockets; a plurality of adhesive vent ports in communication with the first and second cable anchoring pockets; and an adhesive injection port positioned between the adhesive vent ports and in communication with the fiber routing and splice containment passage.

14. The splice protection housing of claim 13, further comprising a main body and a cover.

15. The splice protection housing of claim 14, wherein the cover attaches to the main body by a snap-fit.

16. The splice protection housing of claim 13, wherein the housing is adapted to receive adhesive material for encapsulating optical fibers.

17. The splice protection housing of claim 13, wherein the fiber routing and splice containment passage is defined by two inner walls that extend between the cable anchoring pockets.

18. The system of claim 13, wherein the first and second cable anchoring pockets have a first major dimension and a first minor dimension, the major dimension greater than the minor dimension.

19. The system of claim 18, wherein the fiber routing and splice containment passage has a second major dimension and a second minor dimension, the second major dimension greater than the second minor dimension.

20. The system of claim 19, wherein the first major and minor dimensions are transposed 90° relative to the second major and minor dimension.

21. A splice protection housing comprising; a length that extends between a first end and a second

end; first and second cable anchoring pockets respectively at the first and second ends; a fiber routing and splice containment passage extending between the first and second cable anchoring pockets; a first and second adhesive injection port respectively in communication with the first and second cable anchoring pockets; and a first and second recessed groove respectively associated with the first and second adhesive injection ports.

22. The splice protection housing of claim 21, further comprising a plurality of barriers disposed within the fiber routing and splice containment passage configured to retain adhesive within the first and second cable anchoring pockets.

23. The splice protection housing of claim 22, further comprising a main body and a cover configured to form the fiber routing and splice containment passage when mated; and wherein each the plurality of barriers comprise a body portion and a cover portion, wherein the body portion and the cover portion mate to form a barrier of the plurality of barriers when the main body and the cover are mated.

24. The splice protection housing of claim 22, wherein the plurality of barriers comprise at least one barrier between each of the first and second cable anchoring pockets and a central region of the fiber routing and splice containment passage.

25. The splice protection housing of claim 24, wherein the plurality of barriers comprise at least two barriers between each of the first and second cable anchoring pockets and a central region of the fiber routing and splice containment passage.
